

20. (2.75 points)

Given the following information for a property & casualty insurer:

2015 Earnings	\$700,000,000
Book value at December 31, 2015	\$6,000,000,000

Also given the following information for a sample of other insurers:

	Capitalization (\$ billions)	Price to Earnings	Price to Book Value
Life Ins Co 1	20	16.2	1.8
P&C Ins Co 1	15	12.9	1.4
P&C Ins Co 2	9	11.7	1.5
Health Ins Co 1	6	13.4	1.5
P&C Ins Co 3	3	11.1	1.2
P&C Ins Co 4	1	19.0	1.9

a. (1 point)

Estimate the firm's value by incorporating both the price to earnings and price to book value multiples.

b. (0.5 point)

Describe a risk of using market multiples to value a company.

c. (0.5 point)

Describe a way to counter the risk identified in part b. above while still using market multiples.

d. (0.75 point)

Transaction multiples offer an alternative method of valuation to market multiples. Briefly describe one advantage and two weaknesses of using transaction multiples.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 20	
TOTAL POINT VALUE: 2.75	LEARNING OBJECTIVE: B3: Value the equity of a firm using comparative or relative valuation methods based on multiples of selected financial variables obtained from either peer companies or from underlying fundamentals.
SAMPLE ANSWERS	
Part a: 1 point	
<i>Sample Answer 1</i> Wtd-Avg P-E Ratio using P-C Peer Companies=$[(12.9)(15)+(11.7)(9)+(11.1)(3)+(19)(1)] / 28 = 12.54$ Insurer Value=$(12.54)(700M)=8.778$ Billion Wtd-Avg P-BV Ratio using P-C Peer Companies=$[(1.4)(15)+(1.5)(9)+(1.2)(3)+(1.9)(1)] / 28 = 1.43$ Insurer Value=$(1.43)(6B)=8.57$ Billion Avg Insurer Valuation=$(8.778B+8.57B)/2=8.674$ Billion <i>Sample Answer 2</i> use straight avg of P&C Co Avg PE=$(12.9+11.7+11.1+19)/4=13.675$ Avg P-BV=$(1.4+1.5+1.2+1.9)/4=1.5$ P to E= $700M \times 13.675 = 9.5725B$ P to BV = $6B \times 1.5 = 9B$ Avg=9.29 Billion Note: using the median, as opposed to a weighted average or straight average, was also accepted.	
Part b: 0.5 point	
<i>Sample Answer 1</i> Market Multiples use market values which can be volatile depending on the market's outlook at the time <i>Sample Answer 2</i> Risk profiles can vary greatly by P&C company. Due to primary lines written (long vs short tail), types of coverage (excess, primary), etc. So 1 company's ratio may not reflect another's ratios because of the differences. <i>Sample Answer 3</i> Each industry may have different mix of business and leverage ratios, leading to different k, g, so using market multiples may lead to inaccurate results for calculating value of individual firm.	
Part c: 0.5 point	
<i>Sample Answer 1</i> We can use the averages over several months or years in market values to try to get rid of some of this volatility. <i>Sample Answer 2</i> To counter this, identify 'pure players' that operate in only one LOB & use this to value the insurer in pieces based on premium volume.	

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Part d: 0.75 point
<i>Same Answers for the one advantage</i> • Transaction multiples involve negotiations between sophisticated parties making the valuation more meaningful. • Not subject to random market fluctuations, should have been valued by careful analysis. • Transactions done by people/experts involved in the companies > would have best estimate of values.
<i>Same Answers for the disadvantages, any two of which would earn credit</i> • generally m&a buyers overpay for acquired companies • another weakness is IPOs have historically been underpriced according to some research • Control Premiums – firms will often overpay to get control of another firm. • underlying economic assumptions: historical transactions took place in a different economic environment • transaction multiples are using financial info at the time of the transaction thus may not be up-to-date enough. • Trans. multiples usually include some optimism for synergies created by the merger.
EXAMINER'S REPORT
• The candidate was expected to demonstrate fundamental knowledge and application of valuation methods. • Candidates generally scored well on the question, particularly part a. • Candidates struggled most with part d., often confusing market multiples with transaction multiples. • A common error across parts b., c., and d. was providing insufficiently clear and descriptive responses.
Part a
Candidates were expected to: 1. use proper peer companies in the sample 2. calculate an appropriate multiplier for the sample group of companies 3. estimate the firm's value using P-E and P-BV 4. to use <i>both</i> multiples to estimate the firm's value Common errors: • Candidates included Life and/or Health company in the peer group. • Candidates excluded P&C 1 from the peer group. • Candidates did not follow the estimate through the end using <i>both</i> multiples to estimate the firm's value.
Part b
Candidates were expected to EITHER: 1. cite the fluctuation of market prices over time AND how such fluctuations can impact the valuation range of a price-based multiple OR 2. cite the necessary qualities of an appropriate peer group and/or the perils of constructing an inappropriate peer group AND how the inclusion of an inappropriate peer

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can have an undue effect on the leveraged multiple valuation. • Common errors: ○ Candidates would merely say “different” or “similar” with respect to construction of a peer group without giving any description of what “different” or “similar” could mean. ○ Candidates would merely cite β (beta) as the differentiator of inclusion/exclusions. ○ Candidates would cite capital-size as a reason to exclude a company, but Goldfarb does not recommend using capital-size exclusively as a reason to exclude a company. (In fact, Tables 29 and 31 include AIG in the estimation which is ~10x the average size of the other companies in the peer group.)
Part c Candidates were expected to EITHER: 1. propose using a multi-period average of firms’ prices to reduce the impact of market fluctuations on P-E and P-BV multipliers OR 2. propose the pooling of pure play companies and then weight the respective pure play multiples to create a portfolio-wide multiple that is appropriate for the company being valued. Common errors: ○ Candidates proposed expanding the pool of companies outside of peer-industry group (e.g., include Life and Health, or even Banks or Finance as a whole) to smooth fluctuations. ○ Similar to Part b) above, candidates would suggest using “similar” companies no further description of what “similar” could mean. Candidates were not required to use the term “pure play”, but to describe the idea and/or give examples. ○ Candidates would propose pooling companies based merely on β (beta). ○ Candidates did not mention the weighting of pure play multiples to generate an appropriate portfolio multiple.
Part d • Candidates were expected to cite one advantage and two disadvantages of <i>transaction</i> multiples. • Candidates could offer concise critiques of transaction multiples. Candidates could also combine disadvantages into a broader critique. For example, “Studies show IPOs tend to underprice, while M&As tend to overpay.” • Common errors: ○ Instead of discussing transaction multiples, candidates described advantages and disadvantages of <i>market</i> multiples. ○ As the advantage, candidates stated that it was “easier” to generate a valuation using transaction multiples. ○ Candidates wrote that a transaction multiple was more reflective of a true or real market price than a market multiple because it was based on an agreement between two parties. This was not credited as market prices are also determined by buy-sell agreements between buyers and sellers in the stock market on a frequent, individual basis.